Experiment-20

Name: - N.PARNITA **Regno: -** 192525322

Course Code: -MLA0305 **Slot-**B

Title: Reinforcement Learning for Autonomous Navigation / Intelligent Decision-Making

Aim:

To develop a real-world Reinforcement Learning application such as autonomous navigation, game playing, robotic control, resource allocation, or traffic signal optimization and evaluate its performance using suitable reward metrics.

Procedure:

1. Select a real-world application such as autonomous vehicle navigation or traffic signal optimization.
2. Define the environment and state space.
3. Define the possible actions.
4. Design a suitable reward function.
5. Select an appropriate RL algorithm such as DQN or PPO.
6. Train the agent through repeated interactions.
7. Record rewards, success rate, time, and other performance metrics.
8. Test the trained agent under different conditions.
9. Compare the learned policy with a baseline method.
10. Visualize and analyze the final performance.

```
import random
Q={(s,a):0. for s in ['Low','High'] for a in ['Short','Long']}; alpha=.1;gamma=.9
for _ in range(300):
 s=random.choice(['Low','High'])
 for _ in range(20):
  a=random.choice(['Short','Long']) if random.random()<.2 else max(['Short','Long'],key=lambda x:Q[(s,x)])
  r=5 if s=='High' and a=='Long' else (-3 if s=='High' else (3 if a=='Short' else 1)); ns=random.choice(['Low','High']); Q[(s,a)]+=alpha*(r+gamma*max(Q[(ns,x)] for x in ['Sh
for s in ['Low','High']: print(s,max(['Short','Long'],key=lambda a:Q[(s,a)]),Q[(s,'Short')],Q[(s,'Long')])
```

```
Low Short 39.37065400479568 37.064996275643615
High Long 33.23215180568523 41.06419004898529
```

Result:

The Reinforcement Learning agent successfully learned a policy for the selected real-world application. Its performance was evaluated using cumulative reward, success rate, and other relevant metrics.